

CSMFAB000000000113-Vol-08

Smart Rope Manufacturing for Subsea Tension Applications

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1. Adaptations from CSMFAB000000000022/040 for Marine Use

The standard Smart Rope Umbrella Module (CSMFAB000000000022 V2.0) was designed for high-altitude balloon tether applications. For the AMSR-UBSCLS underwater environment, the following modifications are required:

1.1 Pressure Sealing (New Requirement)

At 100 m operational depth: $P_{\text{external}} = 10.1$ bar Standard 25mm segment bayonet joint gap: 0.1 mm $\rightarrow$ insufficient pressure seal

V1 Marine modification: - Replace PEEK CF40 bayonet with ZrO₂ 3Y-TZP bayonet with O-ring groove - Fluoroelastomer (Viton) O-ring in ZrO₂ groove: rated to 20 bar, -20°C to +200°C - Seawater ingress rate with Viton seal: < 0.001 cc/day (negligible)

1.2 Anti-Corrosion Enhancement

KNbO₃-BaTiO₃ piezo elements: protected by ALD ZrO₂ coating (100 nm, hermetic) LoRa radio module: PEEK CF40 housing + ZrO₂ pressure port UHMWPE fiber: no modification needed (inherently saltwater-immune) ZrB₂-SiC body: no modification needed

1.3 Ballasting for Neutral Buoyancy

Standard 25mm segment mass: 8.6 g Segment volume: 15.6 cm³ Segment buoyancy (seawater): $1.025 * 9.81 * 15.6e-6 = 0.157$ N Segment weight: $8.6e-3 * 9.81 = 0.0844$ N

Segment net buoyancy: $0.157 - 0.0844 = +0.073$ N (**positive buoyancy — floats!**)

This is favorable: the rope will naturally hang upward, maintaining tension toward the surface. For rope neutrally buoyant condition, insert tungsten carbide ballast sphere per segment: WC density = 15,800 kg/m³; needed ballast mass = 7.3 g/segment $\rightarrow$ volume = 0.46 cm³

Design choice: Keep rope positively buoyant — the natural upward tension helps maintain rope geometry during pre-launch setup.

2. Manufacturing Process — Marine Grade Smart Rope

2.1 Segment Fabrication Sequence

Step 1: Flash-sinter ZrB₂-SiC segment cube at 1800°C (CSMFAB000000000001 protocol) Step 2: Machine bayonet features with O-ring groove using diamond tooling Step 3: ALD-coat KNbO₃-BaTiO₃ actuator with 100 nm ZrO₂ (atomic layer deposition system) Step 4: Install KNbO₃-BaTiO₃ bimorph into ZrB₂-SiC body under argon atmosphere Step 5: Thread UHMWPE Dyneema SK99 through segment fiber channels (pre-tensioned to 2% strain) Step 6: Install LoRa micro-module into PEEK housing; pot with marine epoxy Step 7: Install Ti₃AlC₂ MAX Phase hinges with MXene-compatible lubricant Step 8: Hydrostatic test: 20 bar (200 m equivalent), 1 hour — zero leakage required Step 9: Functional test: deploy/retract umbrella mechanism 50x under 10 bar pressure

2.2 Pre-Tensioning Protocol

UHMWPE fiber must be pre-tensioned to store elastic energy:

1. Thread fiber through assembled segment chain
2. Connect to tensioning rig (hydraulic, 0-5000 N range)

3. Apply tension to 50% of $F_{break} = 0.5 * 30 \text{ strands} * 3336 \text{ N} = 50,040 \text{ N}$
4. Hold for 5 minutes (creep stabilization)
5. Install ceramic end termination (ZrO₂ swaged ferrule)
6. Release tension: fiber settles at ~2% pre-strain → $E_{stored} = 94 \text{ J per 50m rope}$ (see Vol 02)

2.3 Deployment Coil and Underwater Handling

The Smart Rope is coiled on a BFRP spool (diameter 0.8 m) mounted on the stern of each vessel. Deployment: 1. Spear is positioned at depth d (held by pneumatic guide weight) 2. Diver (or ROV) clips rope attachment to spear base in <2 minutes 3. Vessel moves to position on hexagonal ring (GPS-free: LoRa ranging to master vessel) 4. Rope allowed to hang naturally (positively buoyant — rises toward surface plane) 5. Final tension adjustment: motor winch on vessel takes up slack to achieve pre-tension

Deployment time estimate: 20-30 minutes for $n=6$ vessel setup in calm conditions.

3. Rope Lifecycle and Phoenix Protocol

Marine environment expected rope lifetime: - UHMWPE UV exposure (surface portion): stabilizer additive gives >5 year life - ZrB₂-SiC segment durability: indefinite (ceramic, no corrosion mechanism) - KNbO₃-BaTiO₃ actuator cycles: rated >10⁸ cycles at 50% max displacement

Post-launch recovery protocol: 1. Rope automatically floats to surface after spear release (positive buoyancy) 2. BFRP collection ring (orange-colored YInMn + CoAl₂O₄ high-vis pigment) aids visual recovery 3. Rope coiled onto vessel spool for inspection and reuse 4. Phoenix Protocol: any segments with >20% actuator degradation → sent to Aegis-Forge I for recovery

Recyclability: - ZrB₂-SiC segments: re-SPS after grinding → 94% property retention - UHMWPE fiber: biodegrades over 10-50 years in ocean (bio-based Braskem variant: 3-5 years) - KNbO₃-BaTiO₃: HNO₃ dissolution → 97% elemental K, Nb, Ba, Ti recovery

Citations (Vol 08)

- CSMFAB000000000022 V2.0, Segment manufacturing protocol, CSM 2026
- CSMFAB000000000040 V2.0, Production bayonet design, CSM 2026
- DSM Dyneema, SK99 Technical Datasheet, 2025
- Viton O-Ring technical manual, Parker Hannifin, 2024
- ASTM F1045-21, “Performance Requirements for High Modulus, Ultra-High Molecular Weight Polyethylene Fiber”
- ALD process: Leskelä, M., Ritala, M., “Atomic Layer Deposition,” Angew. Chemie 2003

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